

24th March 2026

Individual Assignment

Student

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Subject

Software Architecture



Context

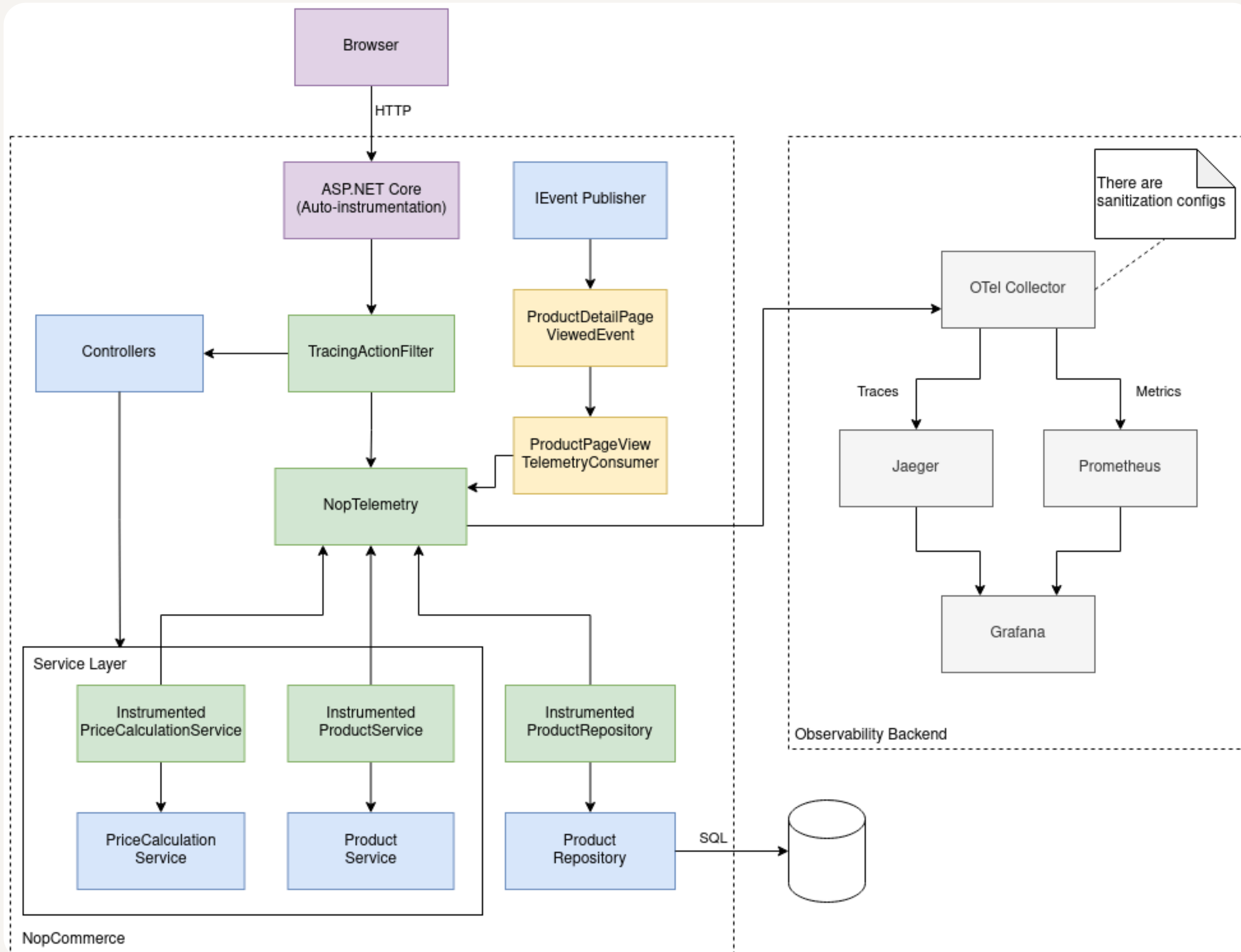
01 nopCommerce
Production-grade open-source e-commerce platform, ASP.NET Core, layered architecture

02 No built-in observability
No OpenTelemetry, no tracing, no custom metrics out of the box

03 Goal
Instrument the "Customer searches and views a product" flow end-to-end

04 Constraint
Zero changes to business logic; all instrumentation in the infrastructure layer

05 Stack
OpenTelemetry SDK → OTel Collector → Jaeger (traces) + Prometheus (metrics) → Grafana



Custom Metrics

01 Search Duration
Tracks search query latency to detect performance degradation in the search pipeline

02 Empty Results
Monitors zero-result queries to identify catalog completeness issues or search index failures

03 Top Searched Keywords
Ranks most frequent search terms for business intelligence and catalog gap analysis

04 Pricing & Repository Duration
Measures operation latency at the pricing and data access layers to isolate bottlenecks

infra/otel-collector/config.yaml

```
processors:  
  attributes/sanitize:  
    actions:  
      - key: client.address  
        action: delete  
      - key: net.peer.ip  
        action: delete  
      - key: http.request.header.cookie  
        action: delete
```


Demonstration